

# Segmenting Minerals from a Rotating Thin Section

*The Moon is my beautiful impossibility.*

Ping Wang

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Tutorial • Colab Notebooks • Lunar Petrographic Exhibit

This guide introduces how to turn a rotation series of thin section photomicrographs into a mineral map, using the Phase Segmentation app (<https://huggingface.co/spaces/pingwang2026/phase-segmentation>). You will identify the major four rock-forming minerals (opaque, plagioclase, pyroxene, and olivine) and glass in a lunar rock, separate them by phases, and export a short animation of each mineral. No coding is required.

## 1 Basic Concepts

### 1.1 Thin sections

A thin section is a slice of rock ground down to about 30 micrometres (0.03 mm) and glued to a glass slide. At that thickness most rock-forming minerals become transparent, so light shines through them and we can study how each mineral bends and polarizes that light. The 30  $\mu\text{m}$  standard matters because the colours you see under crossed polars are calibrated to it.

### 1.2 PPL & XPL

The polarizing microscope has two polarizing filters, and you switch between two setups:

*Plane-polarized light (PPL)* uses only the lower polarizer. Here you examine a mineral by its natural colour, its relief (how strongly its outline stands out, which reflects its refractive index), its cleavage (planes of weakness that show as straight cracks), and its shape.

*Cross-polarized light (XPL)* adds a second polarizer rotated  $90^\circ$  to the first (hence "crossed"). Now you see interference colours produced by the mineral splitting light into two rays that travel at different speeds. The strength of this effect is called birefringence, and different minerals show characteristically different interference colours (compared against the Michel-Lévy colour chart).

### 1.3 Extinction: Why Use Rotational Series

When you rotate the stage under crossed polars, an ordinary crystalline (anisotropic) mineral goes dark four times per full turn, once every  $90^\circ$ , and shows its bright interference colour in between. Some materials, though, stay dark at every angle:

**Isotropic** materials (volcanic glass, and cubic minerals) never produce interference colours, so they are always dark in XPL — but they still transmit light, so they look clear and bright in PPL.

**Opaque** minerals (metal oxides such as ilmenite) are dark in both PPL and XPL, because no light passes through them at all.

Here is the catch that makes a single PPL/XPL image insufficient: a normal mineral that happens to be at extinction in one frame also looks dark in that frame. You cannot tell it apart from truly always-dark material from one image.

By photographing the section at many rotation angles, you can see which grains are dark at every angle (opaque or glass) and which are only momentarily dark (crystalline minerals at extinction). That is exactly why the input is a 36-frame rotation series, and it is the principle the app's automatic steps rely on.